Mikhail Balkhaev

ML Engineer / Generative AI Engineer / Product Developer

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ML engineer working on generative AI: image and video diffusion, speech, LLM agents, search and the GPU infrastructure behind them. I take products from the model pipeline and quality evaluation to the backend, the interface and production. Embeddings and vector search (Qdrant, pgvector) with hybrid retrieval and reranking run through most of my projects: companion memory in my own platforms, RAG at Funcorp, game search on ggpick.club, the Polywin research platform and a sales CRM. Before that, 13 years of web development, including VK (Delivery Club) and Yandex (Yandex Eats).

Experience

Entrepreneurship, luv.club, adorely.co, sinly.io — Mar 2026 – present

Founder, Generative AI Engineer

- Founded and built my own AI companion platforms (18+) with image, video and 3D generation, chat and voice: a multi-tenant Hono backend, Redis/BullMQ workers and RunPod GPU jobs with idempotent submission.
- Identity-consistent photos with Qwen Image 2.1: an edit-consistency LoRA, up to four reference images and an automatic redraw when the face drifts.
- Video: text- and image-to-video with two-pass sampling and RIFE interpolation to 48 fps; face and full-character swap with MiniMax H3 and swap LoRAs.
- 3D stage to video: the reference render conditions only the first 5 of 20 sampling steps, which keeps the camera and composition while the model produces natural motion.
- Companion memory: LLM fact extraction with corrections, hybrid retrieval over OpenAI embeddings in pgvector, lexical reranking and LLM relevance scoring.
- Age-safety checks for generated media: face detection, a ViT age classifier and a logistic probe on CLIP ViT-B/16 embeddings trained on 9,574 labelled person crops. On held-out data, correct refusals rose from 283 to 350 of 400 and false refusals fell from 26 to 3 of 302.
- Tool-using Grok agents: a studio director that sets up shoots and reviews results, and a 3D scene editor with validated, revision-checked edits; REST and MCP API with OAuth 2.1.
- Voice: self-hosted Kokoro-82M TTS on CPU (ONNX, 54 voices) and Whisper recognition.

Funcorp, Cyprus — Nov 2024 – Mar 2026

Senior Product Developer

- Started on the main product, then built new AI products from scratch.
- Diffusion pipelines for images and video (Wan 2.2, LTX, SD, SDXL) in ComfyUI and DiffSynth-Studio; dataset collection and LoRA training.
- Text-to-speech: voice design and fine-tuning the model for natural-sounding speech.
- LLM features with Qwen3.5-9B, ChatGPT and Grok; RAG on Qdrant.
- Data and backend: Kafka, ClickHouse, PostgreSQL, Redis, Next.js, Hono, FFmpeg.

AI and media

Studio, film.balkhaev.com

AI film production: a Claude producer agent turns an idea into a script and staged shots, renders reference passes in Blender over MCP and generates the shots with H3. A take is accepted only after frame decoding, a vision-model review and a speech-transcription check.

MediaPipes, mediapipes.balkhaev.com

Image, video and speech generation on ComfyUI and vLLM: Qwen Image 2.1 INT8 with up to ten references, tiled upscaling to 8 MP, Higgs TTS 3 (4B). INT8 attention for H3 cut sampling from 13.0 to 6.8 s/step (a swap job from 129 to 80 s) and shipped after identity, lip-sync and LPIPS metrics and a blind review.

Persons

Character identity service: seven reference views and a character sheet per character, CPU face checks with YuNet and 512-d AuraFace embeddings, storyboard stills with up to four characters, and a PostgreSQL job queue with SKIP LOCKED and per-person fairness.

Modeler, modeler.balkhaev.com
Characters and previz in Blender: image-to-3D with TRELLIS.2 and Pixal3D to textured GLB, text-to-motion on a 63-bone rig with foot-contact quality gates, Qwen-guided facial reconstruction.

Montage
Video understanding: Qwen3-VL-4B frame descriptions, word-aligned WhisperX transcripts and PySceneDetect cuts in one timeline, with models loaded in turn to fit one GPU; video-to-video recasting with H3.

Inference
GPU scheduler for a home RTX 5080 shared by ComfyUI, local LLMs and Blender: admission by measured RAM and VRAM peaks, GPU sharing only after three comparable solo runs, warm-model residency and budgets that grow after out-of-memory errors.

Local LLM
Qwen3.8-Flash-Next (125B GGUF) on one RTX 5080 through Strata, with experts spread across VRAM, RAM and SSD, a 262k context and about 100 tokens/s; an OpenAI- and Anthropic-compatible gateway with multi-account routing.

Client projects

ZemStroy CRM
Sales CRM for a land development company, designed and built from scratch: leads from amoCRM, Avito and CIAN in one pipeline on Redis/BullMQ queues; every call transcribed with Whisper and scored by an LLM; a first-response bot with versioned prompts; semantic search on embeddings in Qdrant; Telegram alerts on new leads and SLA breaches.

Altai Land, алтайземля.рф
Sales website for a land development project in Altai with an interactive 3D valley on Three.js: scene packages streamed to mobile devices (first load cut from 7.2 to 3.7 MB), a chunked forest of 79,125 trees (5.9M to 1.2M triangles), a 568-plot masterplan shared by the 2D and 3D views, and 31 indexable pages for search traffic.

Data and search products

ggpick.club
Multilingual game discovery service in Russian, English and Chinese: Steam and community data in PostgreSQL, hybrid search on multilingual-e5 embeddings in Qdrant with cross-encoder reranking and graceful degradation when models are unavailable, transliteration and keyboard-layout correction for queries. 200+ unique visitors a day.

Polywin
Research platform for Polymarket prediction markets: typed market-data clients with WebSocket order-book reconciliation, an opportunity scanner that accounts for book depth and fees, semantic retrieval on embeddings in Qdrant and LLM analysis of news with validated quotes, forecasts scored by Brier score and calibration. A strategy emits signals only after 30+ resolved markets with a statistically significant edge.

Web development

Yandex, Moscow — Oct 2022 – Aug 2024
Frontend Developer

- Moved from VK with the Delivery Club team to the Yandex Eats advertising unit: analytics and fast testing of ad hypotheses.
- Main Yandex Eats and Yandex Vendor webviews; micro-frontend architecture.
- An ad runner game whose first partner was Haval at its Russian launch; most of the banner ads, analytics debugging and new metrics.

VK, Delivery Club — Jun 2020 – Oct 2022
Lead Developer

- Maintained delivery-club.ru, effectively full-stack: admin panels for partners and lawyers, ad formats, internal UI and API libraries.
- After an incident, built a document tool for the legal team with versioning and audit logs.

Data-screen, Moscow — Jun 2019 – Mar 2020
Full-stack Developer

- A sports streaming service built from high-level ideas with full technical ownership (Node.js, React, PostgreSQL, Redis, Docker); COVID put it on hold.

Smart Solution, Moscow — Nov 2015 – Dec 2018
Senior Frontend Developer

- Frontend for about ten high-traffic sites in the Alexa top 50: legacy rewrites, optimisation and builds; ad integrations; trained developers and reviewed code.

Oversun, Web Developer — 2014 – 2015
WordPress sites from scratch with custom modules; led a junior developer on the company website.

Ideegroupe, Sochi, Web Developer — 2013
WordPress and ModX: modules for a city news portal, SQL optimisation.

RedBox, Buryatia, HTML Developer — 2011 – 2012
Joomla and WordPress templates, jQuery.

Skills

Generative models	Qwen Image, H3, Wan 2.2, LTX, SD, SDXL, TRELLIS.2, Kokoro, Higgs TTS, Whisper, Qwen3-VL
ML engineering	LoRA training, datasets, evaluation (identity similarity, LPIPS, lip-sync, held-out sets), CLIP probes, embeddings and RAG, INT8 and GGUF inference
Serving	ComfyUI, vLLM, ONNX Runtime, RunPod, GPU scheduling, Redis/BullMQ
Search and RAG	Embeddings (OpenAI, multilingual-e5, CLIP, AuraFace), Qdrant, pgvector, hybrid search, cross-encoder reranking, RAG, LLM memory
LLM and agents	Tool use, MCP, Claude, GPT, Grok, Qwen
Backend and data	Python, TypeScript, Node.js, Bun, Hono, FastAPI, PostgreSQL, ClickHouse, Kafka, Redis, Qdrant, SQLite, Docker
Frontend and 3D	React, Next.js, Three.js, WebGL, Blender, Playwright

Details

Languages	Russian (native), English (upper intermediate, B2)
Education	Secondary education
Work	Full-time, part-time or project work; remote, hybrid or on-site; open to business travel and relocation within Russia.